Swami Agnivesh Shaga

Final-Year Undergraduate
Computer Science & Engineering (AI & ML)

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Examination	Degree/Certification	Institute	Year	CPI/%
Graduation	B.Tech. (CSE–AIML)	**VNR Vignana Jyothi Institute of Engg. & Technology**	2023–2027	**8.55**/10.0
Intermediate	BIE (XII), MPC	Acumen Junior College, Warangal	2022	91.6%
Matriculation	SSC (X)	Shine High School, Warangal	2020	10.0/10.0

Work Experience

Alfago Research LLP *Dec 2025 – May 2026*
Software Engineer Intern | Hyderabad

- **Execution Backend:** Engineered a **Python/asyncio** service with **FastAPI** and **AWS SQS** to reconcile account-specific target positions across a **three-account** model in a two-person team
- **Broker Integration:** Integrated **Tradejini** execution and **Fyers** market data with **OAuth 2.0**, canonical instrument mapping, per-account rate limiting and static IP routing through **AWS Elastic IPs**
- **Reliability & Risk:** Established duplicate-order protection with deterministic IDs and **PostgreSQL** constraints; resolved ambiguous timeouts through an **order state machine**, sequenced multi-leg fills and added scoped **kill switches**
- **Operations Dashboard:** Developed **React/JavaScript** views for positions, P&L, account sessions and audit history, streaming prices through **Server-Sent Events** with per-account **paper/live** controls
- **Cloud & Testing:** Provisioned **EC2/RDS**, **Secrets Manager**, **SNS**, **Nginx** and **systemd**; persisted transactional audit trails and verified execution with **127 automated tests**

Skills

- **Languages:** C, C++, Python, Java, JavaScript, R, SQL
- **ML & Data:** scikit-learn, TensorFlow, Pandas, NumPy, SHAP, RAG
- **Backend & Web:** FastAPI, Flask, asyncio, Node.js, Express, React, PostgreSQL/asyncpg, MongoDB, MySQL, REST APIs, OAuth 2.0, SSE, WebSockets
- **Cloud & DevOps:** AWS (EC2, RDS, SQS, S3, Lambda, Secrets Manager, SNS), Docker, Nginx, systemd, GitHub Actions, Render, Cloudflare, Git, Linux, pfSense, OpenVPN
- **Coding Profiles:** LeetCode **1595**, CodeChef **1567** (2 stars), Codeforces **1157**

Achievements

- Ranked **313th of approx. 4,300 teams** in **Amazon ML Challenge 2026**; **1st among VNR VJIET teams**
- Achieved a **Top 10** rank among **300+ teams** at **CodeVerse** (Turing Hut), qualifying as a **Top 25 finalist**
- Placed among the **Top 25** of **400+ teams** at **Nexovate 2025**, Presidency University, Bengaluru
- Earned **50/50** in the **SEBI Investor Certification** covering equity, mutual funds and financial markets
- Maintained a **1-year daily streak** on **LeetCode**, solving **450+ problems** in data structures and algorithms
- Qualified for **Round 2 of AMTI**; placed with distinction in the **SIMO** & **SICHO** olympiads

Key Projects

Automated OHLCV Data Pipeline *Feb 2025*
Python, asyncio, FastAPI, AWS EC2/S3/Lambda, Parquet

- Orchestrated an **AWS EC2** pipeline retrieving **1-minute OHLCV** data for NSE-listed companies, with cron scheduling and **Lambda** for automatic access token rotation
- Implemented asynchronous **token-bucket rate limiting** to prevent API throttling during concurrent data retrieval
- Converted JSON into compressed **Parquet on S3** and served historical data through **FastAPI** for backtesting

Niyantrana *Sep 2025 – Sep 2026*
Metabolic Risk Screening | scikit-learn, SHAP, FastAPI, React, Node.js, MongoDB, Docker

- Modeled **metabolic risk** from **14-day** wearable and diet logs using gradient boosting over **17,961 NHANES adults**; predicted TG, GGT, HbA1c and BP with the Fatty Liver Index and three **isotonic-calibrated** heads
- Diagnosed overlapping-window leakage behind an inflated **R^2 of 0.97**; participant splits performed below a mean baseline. Revalidated on a held-out survey cycle at **AUROC 0.80** and **90%+ sensitivity**, with **SHAP** and monotonic constraints
- Deployed **React 18/Express/MongoDB** and a **145 MiB FastAPI** container (**54–66 ms**) with explicit model failure responses, Samsung Health ingestion, **RAG over 1,014 foods**, **242 tests** and **INR 0/month** hosting

Enterprise-Grade Homelab Network *Dec 2025 – Jan 2026*
pfSense, OpenVPN, VLANs, PKI, Unbound DNS, Omada SDN

- Segmented critical assets, IoT and guests with **802.1Q VLANs** and firewall rules blocking lateral movement
- Secured remote access with **OpenVPN**, an internal **certificate authority** and dynamic DNS; tuned keepalive for carrier NAT timeouts
- Enforced local DNS resolution through **NAT redirection** and added **pfBlockerNG** for ad blocking and IP reputation filtering

Leadership & Activities

Student Volunteer | VNRVJIET
Krithomedh Club, Scintillate Photography Club, ISTE Chapter

- Supported a **24-hour hackathon** for **18+ hours**, mentoring teams, resolving technical issues and managing on-ground logistics
- Contributed to activities of the **Scintillate Photography Club** and the **ISTE Students Chapter**

Independent Algorithmic Trader *Ongoing*
Personal equity & commodities portfolio

- Manage a **INR 340,000** equity and commodities portfolio, applying disciplined **risk management** and quantitative strategies
- Installed a home **IP camera surveillance network**, from cabling to secure remote access